



Tutorix: An AI Toolkit for Bangladeshi Teachers and Students, Grounded in Their Own Textbooks

An AI exercise book for Bangladeshi classrooms: 20 tools for teachers and students, in English and বাংলা, that answer from the user's own textbook using Retrieval-Augmented and Cache-Augmented Generation.

Scientific field	Robotics & AI
Built by	Md. Tahsan Islam, Md. Tanjim Alam Tasin, Shadiya Zaman Tanha and Mufrid Johanee
Supervisor	Md. Ashrafuzzaman

1. Introduction and problem statement

A typical secondary school teacher in Bangladesh teaches several sections with large class sizes. Outside class hours they prepare lesson plans, set and mark question papers, write answer keys and rubrics, and write report card comments for every student. This preparation takes many hours each week and leaves little time for individual support. The national curriculum is also moving towards competency-based learning, which asks teachers to assess higher-order thinking, something that is hard to do consistently without support.

Students face a different problem. Many, especially in rural areas and low-income families, cannot afford private coaching and have nobody to explain a difficult topic, check their answers or help them plan revision before exams. General AI chatbots can help, but they often invent facts, ignore the national syllabus and textbook, handle Bangla poorly, and are not organised around real classroom tasks.

Problem: teachers need to prepare accurate, curriculum-aligned material much faster, and students need a trustworthy personal tutor that follows their own textbook and works in their own language.

2. Project abstract

Teachers in Bangladesh spend many hours each week preparing lesson plans, question papers, answer keys, rubrics and report card comments, while many students, especially outside big cities, have no tutor to explain difficult topics or guide their revision. Tutorix is a web platform that addresses both problems with 20 AI tools, 10 for teachers and 10 for students. Teachers can create lesson plans, Bloom's taxonomy-aligned worksheets, marking schemes, differentiated content and term plans in seconds, and can check test quality with classical item analysis. Students get smart notes, flashcards, practice quizzes with weak-topic diagnosis, mind maps, step-by-step problem solving, concept explanations, mnemonics, study planners and an English-Bangla vocabulary builder. Every tool works in English, Bangla or both. General AI chatbots often invent facts, so Tutorix grounds its answers in the user's own textbook or notes. Short material is read in full through Cache-Augmented Generation, and long material, even a whole book, is searched through Retrieval-Augmented Generation with hybrid keyword and semantic retrieval, so only the relevant passages are used. The system runs on Supabase (PostgreSQL with pgvector, authentication and edge functions) and Groq-hosted open-source language models. All results are saved to a personal library and can be exported as clean PDF or Word documents.

3. Objectives

The main objective of Tutorix is to cut teachers' preparation time by automating lesson plans, question papers, answer keys, rubrics and feedback comments while keeping them aligned with the curriculum. It also aims to give every student a personal AI tutor that explains, quizzes and plans revision in English or Bangla. The project seeks to keep AI answers accurate and syllabus-relevant by

basing them on the user's own textbook or notes through RAG and CAG rather than general internet knowledge. It promotes higher-order thinking by mapping questions to Bloom's taxonomy, and helps teachers improve their assessments with item analysis measures such as difficulty, discrimination, KR-20 reliability and distractor performance. Finally, it aims to stay affordable and fast enough for low-bandwidth, low-budget schools through open-source models and response caching, while protecting user data with role-based access and row-level security so that each user sees only their own work.

4. Proposed solution

Tutorix is a single web application with separate experiences for teachers, students and administrators. Every tool follows the same pattern: the user fills a short form (topic, class, subject, language and optionally their own material), the AI returns a structured result, and the result is saved to the user's library where it can be continued, printed or exported as a PDF or Word document.

4.1 Teacher tools

#	Tool	What it does
1	Bloom's Taxonomy Analyzer	Classifies a question by cognitive level with evidence, or audits a whole paper for Bloom balance with charts and suggested fixes.
2	Lesson Plan Generator	Objectives, timed lesson flow, differentiation, assessment, misconceptions and homework.
3	Rubrics Generator	Analytic, holistic or single-point rubrics whose point weights always add up, with interactive scoring.
4	Question & Worksheet Generator	Six question types across Bloom levels, from a topic or the teacher's own material, with answer key and print layout.
5	Standardized Test Quality Tester	Item flaw detection and rewrites, plus real item analysis from a response CSV: difficulty, discrimination, point-biserial, KR-20, SEM, distractors.
6	Answer Key & Marking Scheme	Model answers, point-by-point marks, acceptable alternatives and common mistakes for any question paper.
7	Differentiated Content Generator	Rewrites a lesson into Support, Core and Extension tiers with scaffolds.
8	Learning Objectives Generator	SMART, measurable objectives tagged by Bloom's level with success criteria.
9	Report Card & Feedback Comments	Unique, specific, growth-focused comments for up to 40 students (CSV import).
10	Syllabus / Term Planner	Spreads the syllabus over real term weeks, skipping holidays, with an assessment calendar and progress tracking.

4.2 Student tools

#	Tool	What it does
1	Smart Notes & Revision Sheets	Outline, Cornell or detailed notes, or a one-page revision sheet of formulas, definitions and exam tips.
2	To-Do List Generator	Breaks a goal into phased, prioritized tasks with time estimates and due dates.
3	Study Planner / Timetable	Day-by-day timetable balanced by exam date, difficulty and confidence, using spaced repetition.
4	Flashcard Generator	Decks from any topic or material, with a flip-and-track study mode.
5	Practice Quiz & Weak Topic Analyzer	AI-generated quizzes with instant grading (including short answers), history, subtopic accuracy and AI diagnosis.
6	Mind Map Generator	Interactive, collapsible concept map with notes on every branch.
7	Concept Explainer	Four explanation levels, analogies, examples, comparisons, self-checks and follow-up chat.
8	Step-by-Step Problem Solver	Math and science solutions in reasoned steps with LaTeX, verification and a practice problem.
9	Mnemonic Generator	Acronyms, acrostic sentences, rhymes, stories and memory-palace images.
10	Vocabulary Builder	English and Bangla meanings, IPA, examples, synonyms, memory tips and a self-quiz.

In addition, **My Library** stores every result with search, favourites and interactive progress (ticked tasks, known flashcards, quiz answers), **My materials** stores textbook chapters and notes for reuse, and the **Admin** area shows usage analytics, manages users and roles, and configures AI models and limits.

5. System architecture and methodology

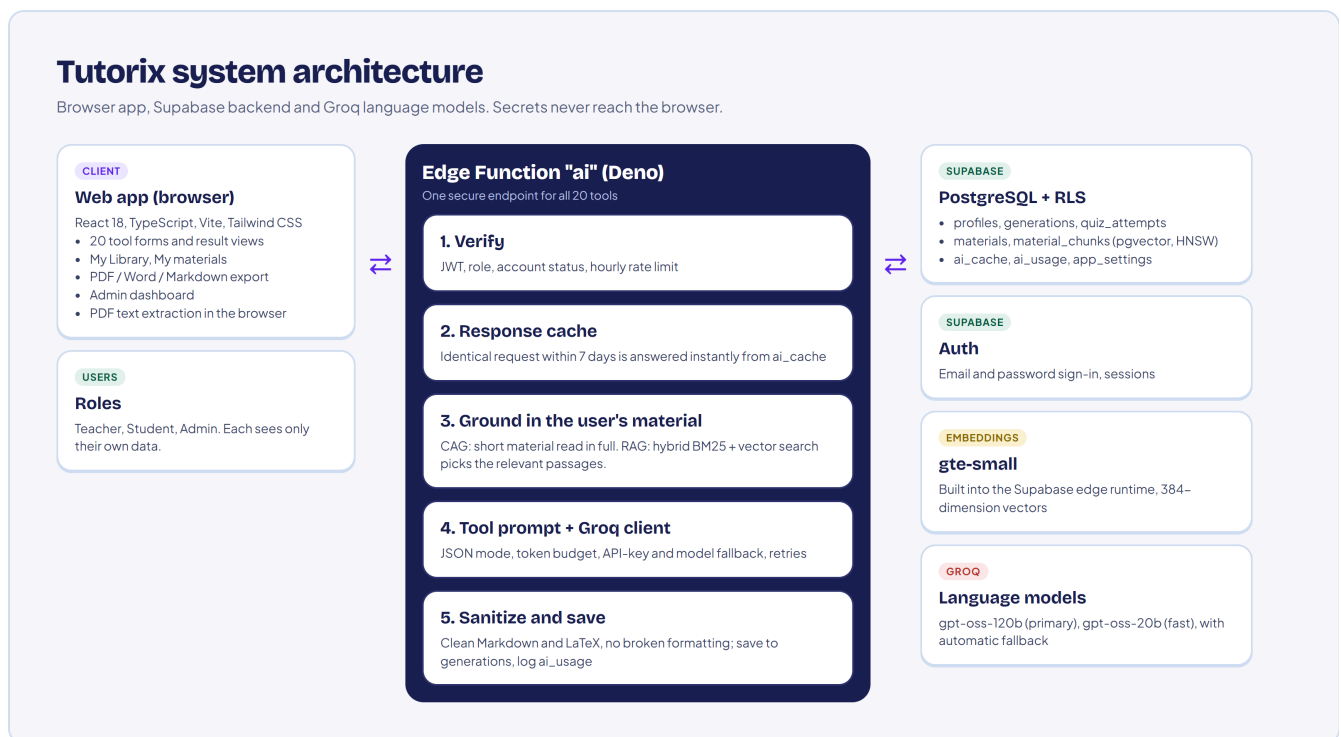


Figure 1. Tutorix system architecture

The frontend is a React 18 and TypeScript single-page application built with Vite and styled with Tailwind CSS. The backend is Supabase: PostgreSQL with row-level security on every table, Supabase Auth for sign-in, and one Deno Edge Function named "ai" that serves all tools. The browser never holds an AI key. For each request the Edge Function verifies the user's session, role, account status and hourly limit, checks the response cache, grounds the request in the user's material, builds the tool-specific prompt, calls a Groq-hosted open-source model in JSON mode, sanitises the output, and saves it. The Groq client keeps every request within the free-tier token budget and automatically switches API key or model when one is rate-limited.

5.1 Grounding answers in the user's own material (CAG and RAG)

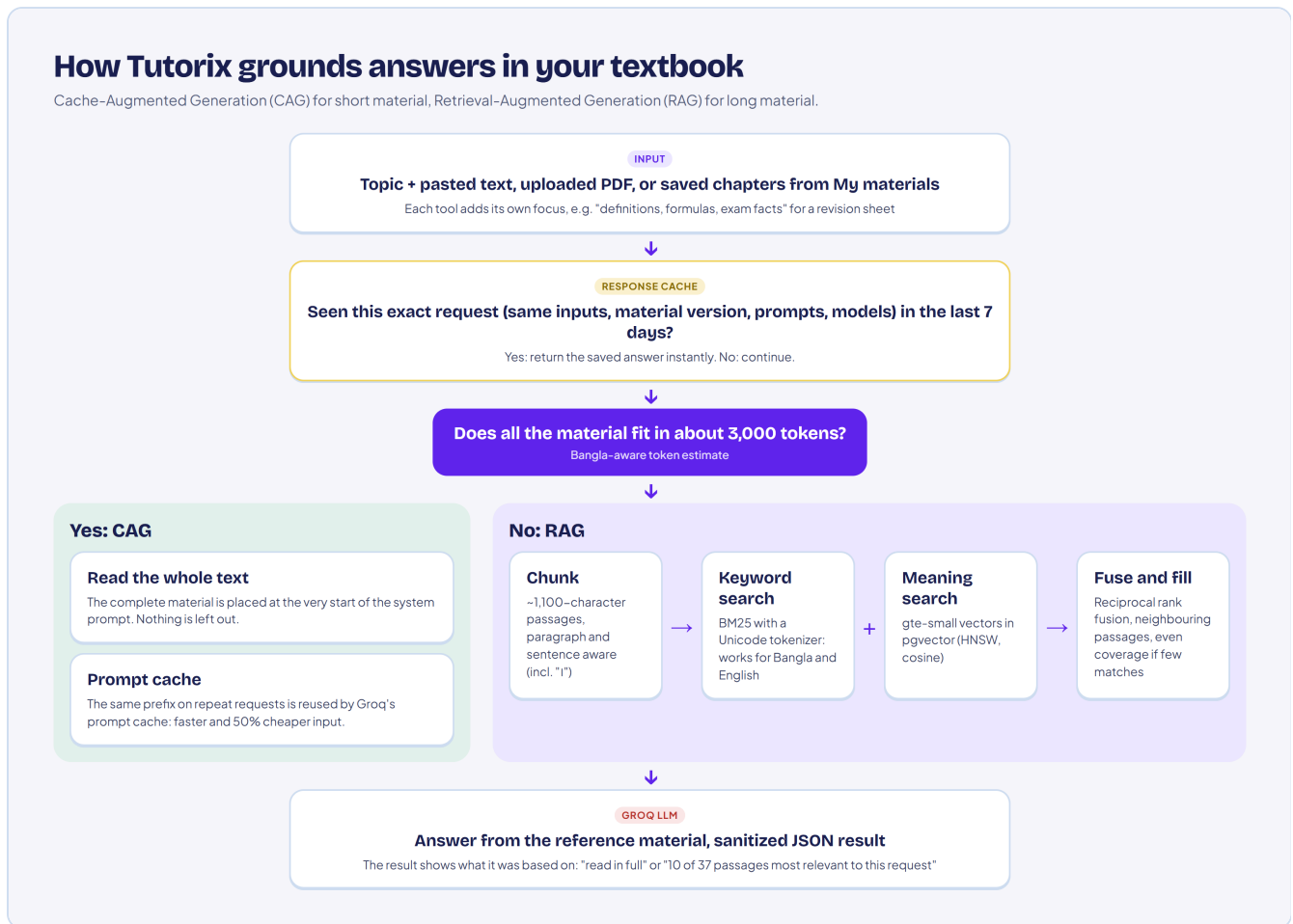


Figure 2. Grounding pipeline: response cache, CAG for short material, hybrid RAG for long material

Cache-Augmented Generation. When the material fits a budget of about 3,000 tokens (roughly 9,000 English characters), the whole text is placed at the very beginning of the system prompt. The model sees everything, and because the prefix is identical across repeated requests on the same material, Groq's prompt cache reuses it, which makes those requests faster and halves the cost of the cached input.

Retrieval-Augmented Generation. Longer material, up to 400,000 characters per item and five items per request, is split into passages of about 1,100 characters that respect paragraphs and sentence endings, including the Bangla "I". Each request builds a query from the topic, subject and a tool-specific focus. Passages are ranked by BM25 keyword search with a Unicode tokenizer that works for Bangla and English, and by semantic similarity using gte-small embeddings stored in pgvector with an HNSW index. The two rankings are merged with reciprocal rank fusion, neighbouring passages are added for context, and if the query matches little, passages are sampled evenly so the whole chapter is represented. Because gte-small is English-only, Bangla passages are handled by keyword search.

Response cache. A SHA-256 key over the tool, the inputs, the version of any saved material, the prompt version and the models identifies identical requests. Within seven days they are answered instantly and do not count towards the hourly limit. "Regenerate" always asks for a new answer, and date-relative tools are never cached.

5.2 Output quality and formatting

All AI text is sanitised twice, on the server and in the browser. The sanitiser repairs LaTeX damaged by JSON escaping, converts stray HTML to Markdown, removes duplicated bullets, code fences and invisible characters, and normalises punctuation. Documents are exported as branded A4 PDFs with rendered math and Bangla text, or as Word files.

5.3 Security and privacy

Row-level security ensures that each user can read only their own generations, quiz attempts and materials. Material passages and embeddings can only be written by the Edge Function, the similarity-search function is callable only by the server, and the response cache is not readable by any user. AI keys and service credentials live only as server-side secrets. Role checks prevent students from using teacher tools and vice versa, and administrators can suspend accounts.

6. Innovation / novelty

The main novelty of Tutorix is that it automatically picks the right way to ground its answers in the user's material. Short material is read in full through Cache-Augmented Generation. Long material goes through Retrieval-Augmented Generation, which combines BM25 keyword search with semantic vector search using reciprocal rank fusion. Most embedding models handle Bangla poorly, so Tutorix matches Bangla text with a Unicode-aware keyword search, which lets Bangla textbooks work as well as English ones. Unlike existing single-purpose apps, it serves both teachers and students on one platform with a shared library and export system. Beyond AI review, its test quality tool calculates real psychometric statistics from student responses, which is rare among AI education tools. It is built for the local context, with Bangladeshi class levels from Class 1 to HSC and O/A Level, bilingual output and Bangladeshi examples. It also stays reliable and low-cost through automatic model and API-key fallback, sanitised output without broken formatting or math, and a seven-day response cache that answers repeated requests instantly at no extra cost.

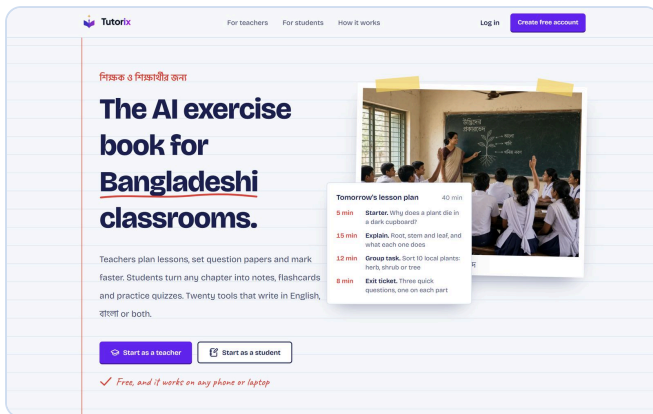
7. Implementation status and evaluation

Tutorix is fully implemented and deployed on Supabase with Groq models. It has been tested end to end with automated scripts that call the live system.

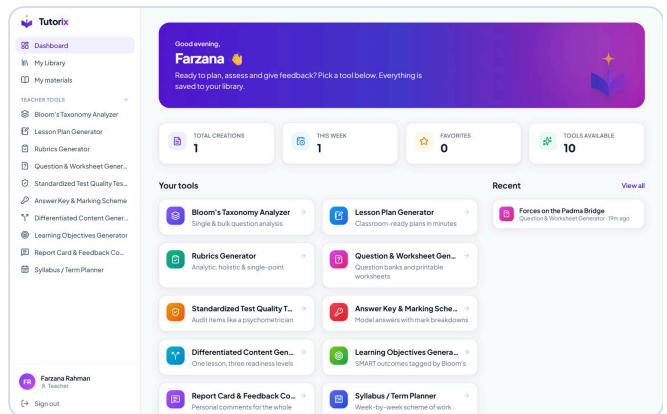
Test suite	What it checks	Result
AI tool tests	All 20 tools plus follow-up, grading and diagnosis endpoints produce valid, well-structured output; role and access checks	25 / 25 passed
RAG / CAG tests	Material upload and indexing, CAG for short text, RAG over saved and pasted English and Bangla text, topic-correct retrieval, duplicate detection, cache hit, cache bypass, cross-user access blocked	13 / 13 passed
Sanitiser tests	LaTeX repair, HTML conversion, tables, currency vs. math, punctuation clean-up	33 / 33 passed

Typical response times are 2 to 7 seconds per tool on the Groq free tier, and repeated requests served from the cache return in about one second. In a retrieval test on a 32,000-character document covering two unrelated topics, a revision-sheet request about Sundarbans tigers retrieved 16 of 37 passages, all from the relevant half, and the generated sheet contained no content from the unrelated topic.

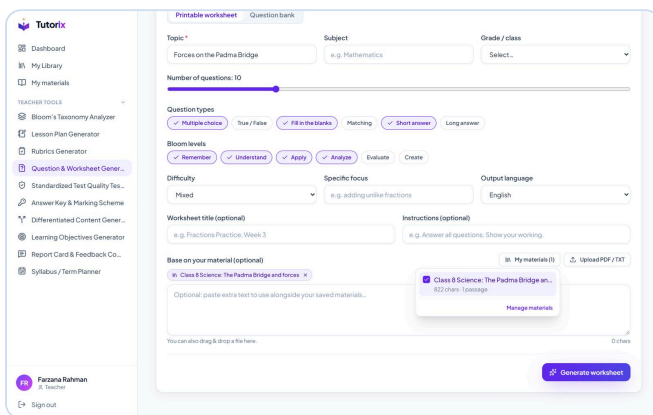
8. Screenshots



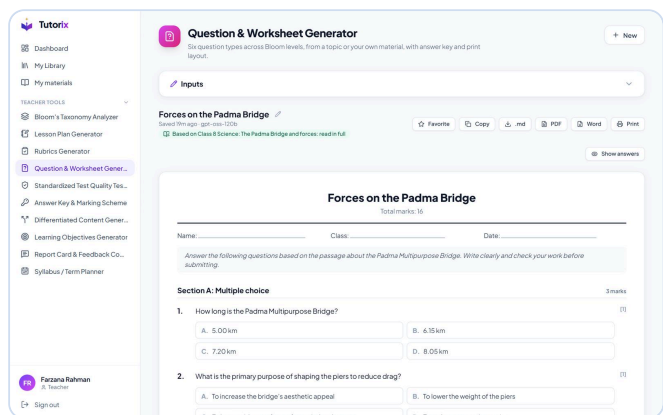
Landing page



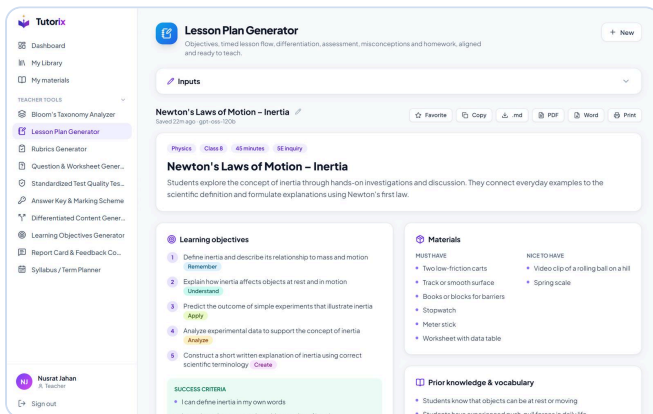
Teacher dashboard



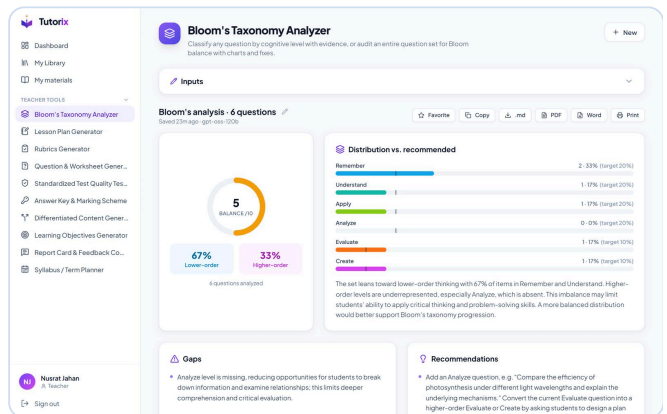
Attaching a saved chapter from My materials



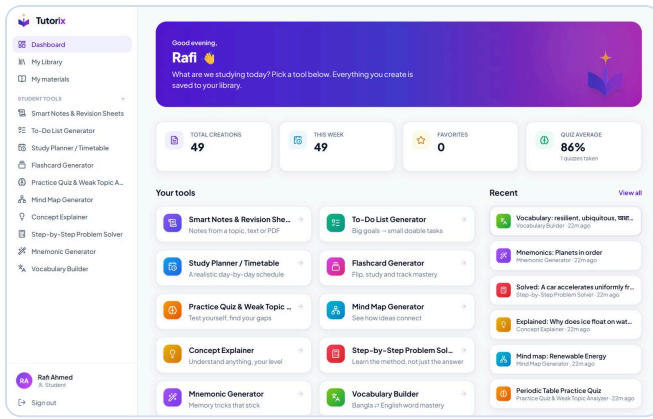
Worksheet grounded in that chapter



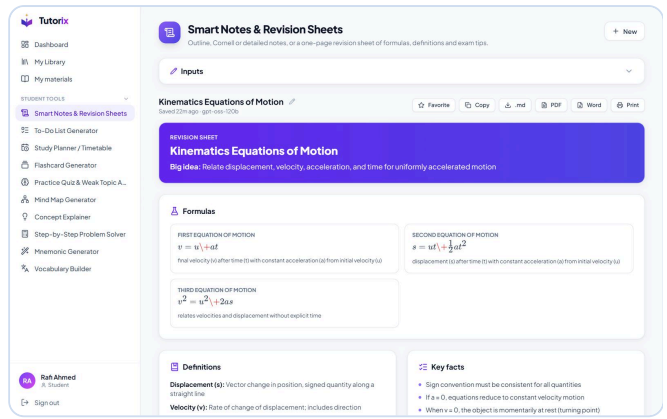
Lesson plan



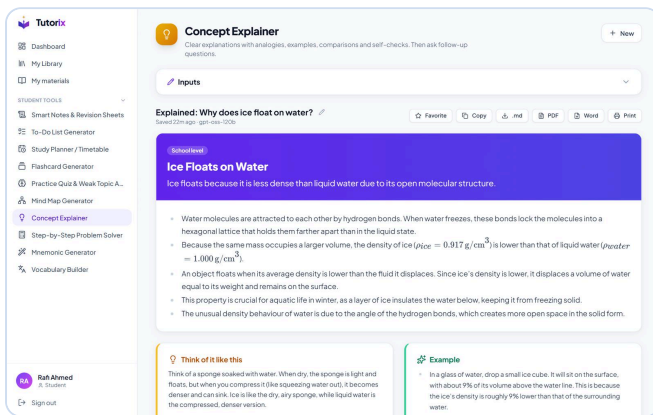
Bloom's taxonomy analyzer



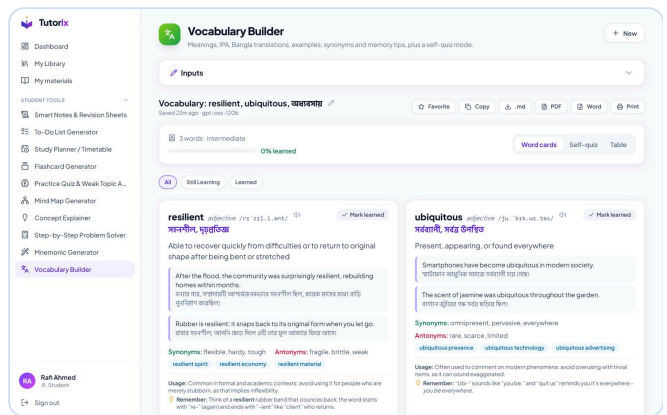
Student dashboard



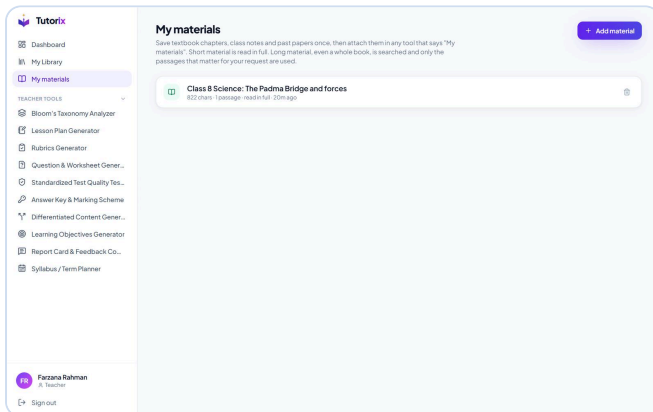
Smart notes



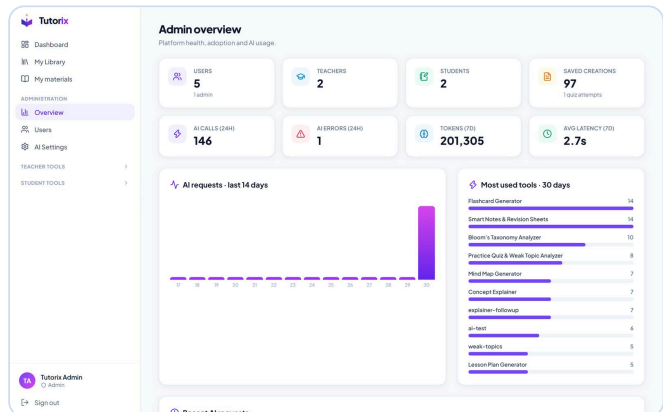
Concept explainer



English and Bangla vocabulary builder



My materials



Admin overview

9. Expected impact

Tutorix is expected to reduce teachers' preparation work from hours to minutes, giving them more time for teaching and individual student support, while making their assessments more balanced across Bloom's levels and more reliable. Students will get free, round-the-clock tutoring in their own language based on their own textbook, which especially helps rural and low-income students who cannot afford private coaching and so reduces the learning gap. Schools will benefit from better question quality and data-driven insight into weak topics, supporting the competency-based approach of Bangladesh's new national curriculum. Because it runs in any web browser on low-cost infrastructure, it can scale to thousands of schools and later expand to more subjects, local exam boards and offline or mobile use. It also has research value, showing how grounding methods like RAG and CAG can make AI trustworthy for education in low-resource languages such as Bangla.

10. Future work

The next steps are a pilot with teachers and students in a few schools to measure time saved and learning gains, a multilingual embedding model so that Bangla material also benefits from semantic search, ready-made libraries of NCTB textbook chapters, an installable mobile version with offline access to saved results, classroom features that let teachers share quizzes and track class-wide weak topics, and optional support for local exam board formats.

11. Team

Tutorix was built by Md. Tahsan Islam, Md. Tanjim Alam Tasin, Shadiya Zaman Tanha and Mufrid Johanee. Our supervisor is Md. Ashrafuzzaman.